

# Software Hackathon 2026

Domain: **Software Development** (Math application)

|              |                    |
|--------------|--------------------|
| Team Name    | LocalHost one      |
| Team Leader  | J V Karthikeyan    |
| Team Members | Dheepak Adhithya S |

Github Repository Link :

<https://github.com/DHEEPAK001/Helix-Math-portal->

Tech Stack:

| Category   | Technology   |
|------------|--------------|
| Frontend   | React        |
| Build Tool | Vite         |
| Styling    | Tailwind css |

|                       |                             |
|-----------------------|-----------------------------|
| Routing               | React Router DOM            |
| HTTP Client           | Axios                       |
| Backend Language      | Java                        |
| Backend Framework     | SpringBoot framework        |
| ORM                   | Spring Data JPA (Hibernate) |
| Security              | Spring Security             |
| Validation            | Spring Validation           |
| Database              | Mysql                       |
| DTO Mapping           | MapStruct                   |
| Boilerplate Reduction | Lombok                      |
| Build Tool            | Maven                       |
| Containerization      | Docker                      |
| Orchestration         | Docker Compose              |
| Web Server            | Nginx                       |
| Version Control       | Git                         |
| Repository Hosting    | Github                      |
| IDE                   | AntiGravity, IntelliJ       |
| AI Integration        | OpenRouter Api              |

## Project Description

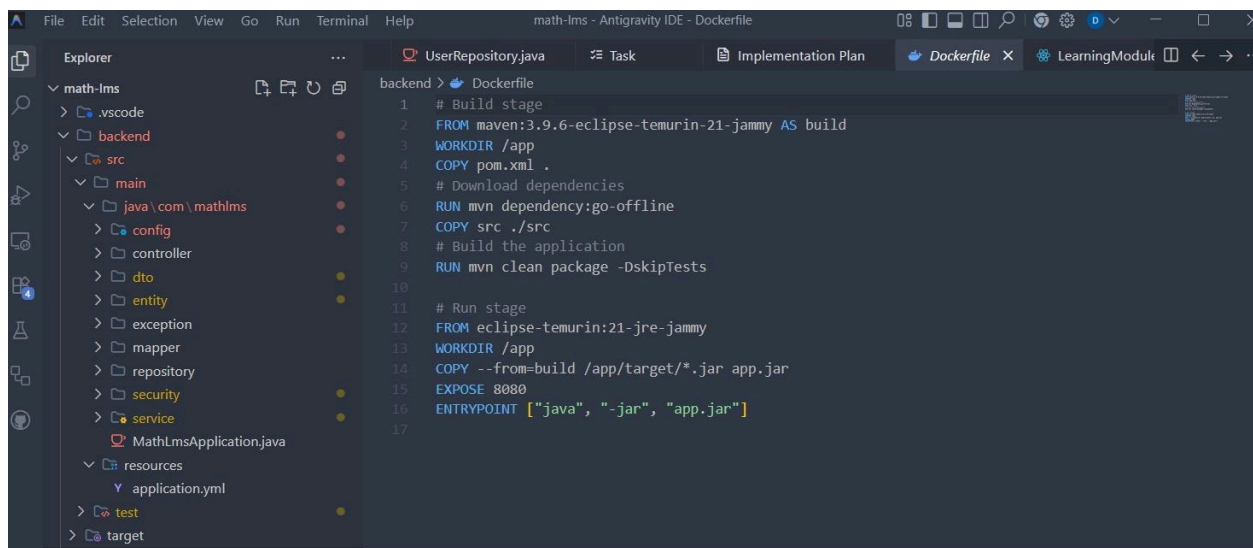
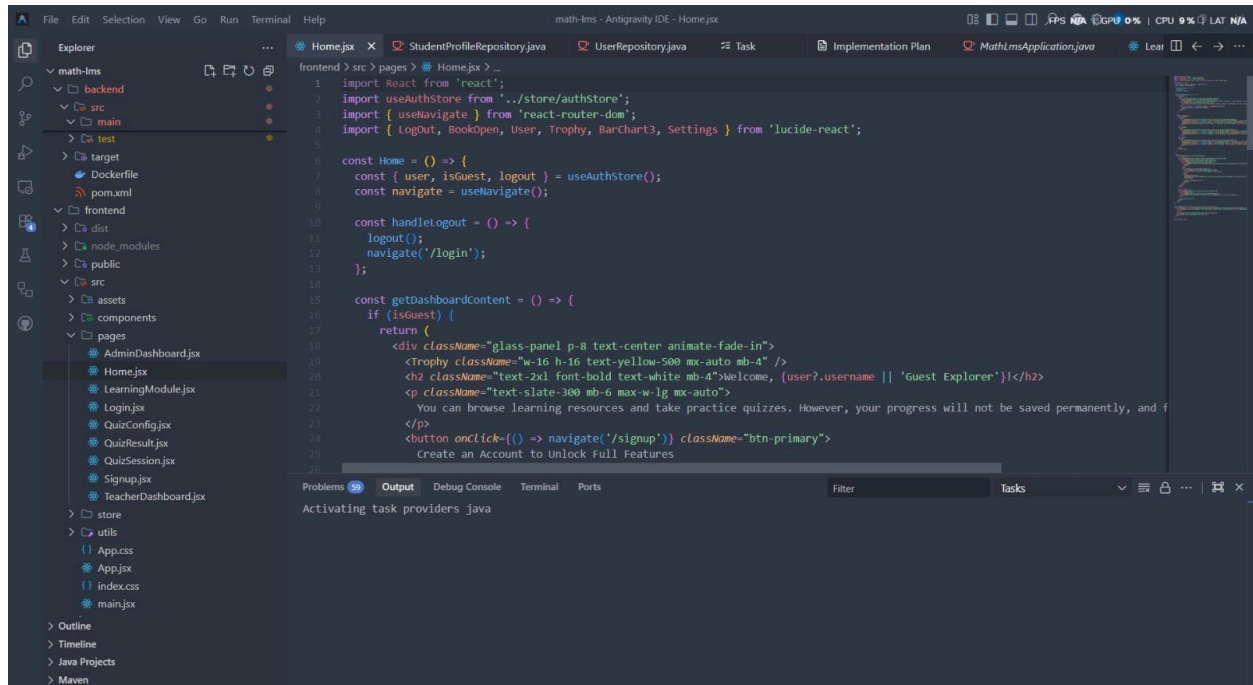
Math LMS is a full-stack Learning Management System designed to provide an interactive and intelligent platform for mathematics education. The application bridges the gap between educators and students by offering digital classrooms, automated assessments, AI-powered assistance, and comprehensive learning analytics.

The platform supports three distinct user roles—Administrator, Teacher, and Student—each with dedicated dashboards and permissions. Teachers can create courses, upload learning materials, manage quizzes, and monitor student performance. Students can access learning modules, participate in quizzes, receive instant feedback, and interact with an AI tutor for personalized assistance. Administrators oversee user management, platform configuration, and system-wide analytics.

The system follows a modern client-server architecture using React for the frontend and Spring Boot for the backend. Authentication is secured through JWT-based authorization, while MySQL serves as the relational database. Docker Compose enables seamless deployment by containerizing the frontend, backend, and database into a unified environment.

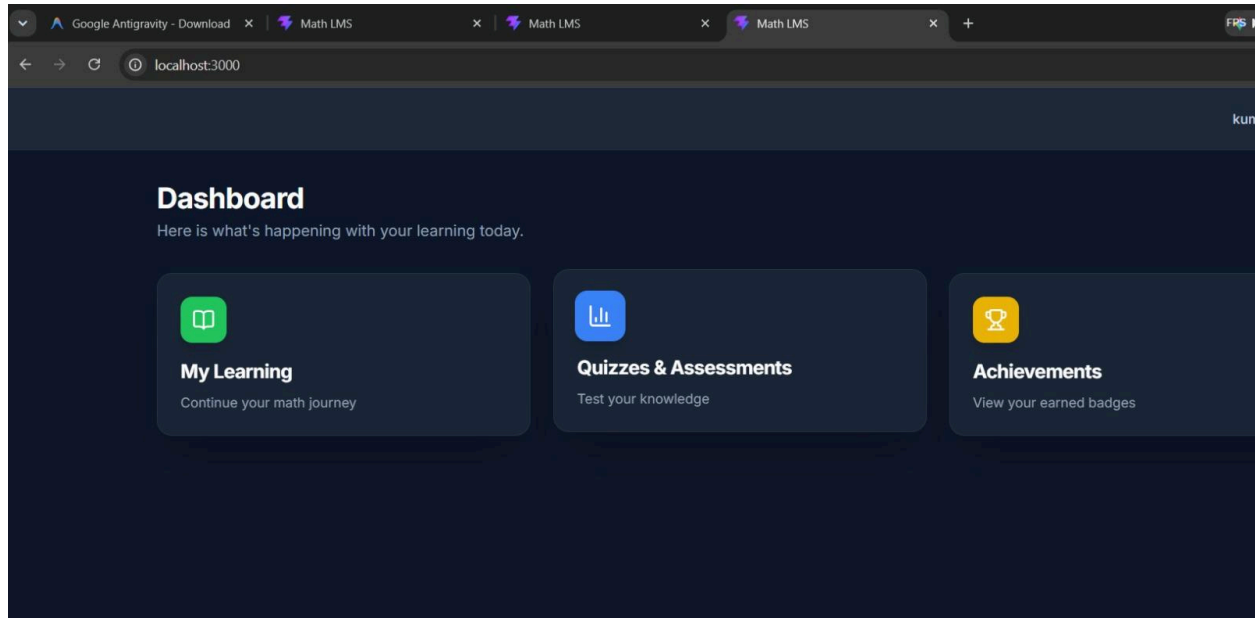
The primary objective of Math LMS is to make mathematics education engaging, scalable, and accessible while reducing the administrative workload for educators through automation and intelligent features.

## Code Images from development stages:





# BANNARI AMMAN INSTITUTE OF TECHNOLOGY & INSTITUTION'S INNOVATION COUNCIL





## Quiz Configuration

Setup your practice environment

Practice Mode

**Assigned  
Assessments**

My Results

### Mock test

skill test

Due: 8/1/2026, 9:29:00 AM

Start



### Mock Test -2

test

Due: 8/1/2026, 10:29:00 AM

Start





## Quiz Configuration

Setup your practice environment

Practice Mode

Assigned  
Assessments

My Results

### Subtraction (EASY)

Wow, a perfect score! You clearly have a fantastic grasp...

100%  
Accuracy

4

### Counting (EASY)

Don't worry—every expert started somewhere, and this ...

0%  
Accuracy

0

**Assessment Session**🕒 14:56🏁 Finish Test

Question 1 of 4

🚩 Flag for review

Which option shows the number 5?

☐ 4

☐ 5

☐ 6

☐ 7

< Previous

Next >

**Question Palette**

1

2

3

4

■ Answered

■ Flagged

■ Unanswered

**Teacher Portal**  
mano@gmail.com

📖 Question Bank

**👤 My Students**

📄 Assessments

📊 Analytics

**My Students**

+ Add Student

D

dheepakadhithya@gmail.com

Assigned recently

🗑

K

kumar@gmail.com

Assigned recently

🗑

J

jagan@gmail.com

Assigned recently

🗑

S

sachin@gmail.com

Assigned recently

🗑



Teacher Portal  
mano@gmail.com

Question Bank

My Students

Assessments

Analytics

Assessments

Create Assessment

Mock test

skill test

Starts: 8/1/2026, 7:30:00 AM Due: 8/1/2026, 9:29:00 AM

Edit Settings

Mock Test -2

test

Starts: 8/1/2026, 8:30:00 AM Due: 8/1/2026, 10:29:00 AM

Edit Settings

Student Results

| Student         | Topic               | Score | Accuracy | AI Feedback                                          |
|-----------------|---------------------|-------|----------|------------------------------------------------------|
| kumar@gmail.com | Subtraction<br>EASY | 4     | 100%     | Wow, a perfect score! You clearly have a fantasti... |
| kumar@gmail.com | Counting<br>EASY    | 0     | 0%       | Don't worry—every expert started somewhere, a...     |

Teacher Portal  
mano@gmail.com

Question Bank

My Students

Assessments

Analytics

My Students

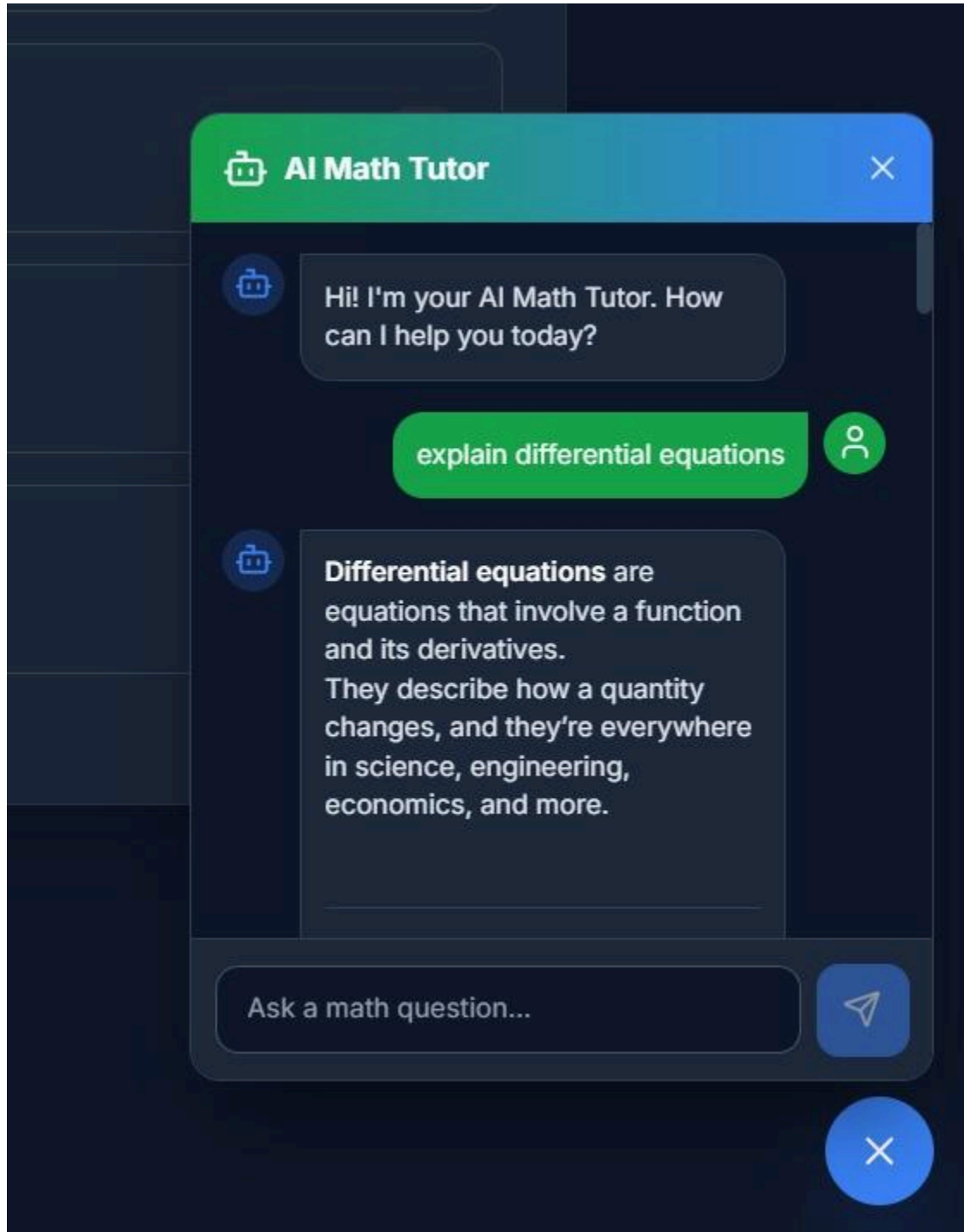
Add Student

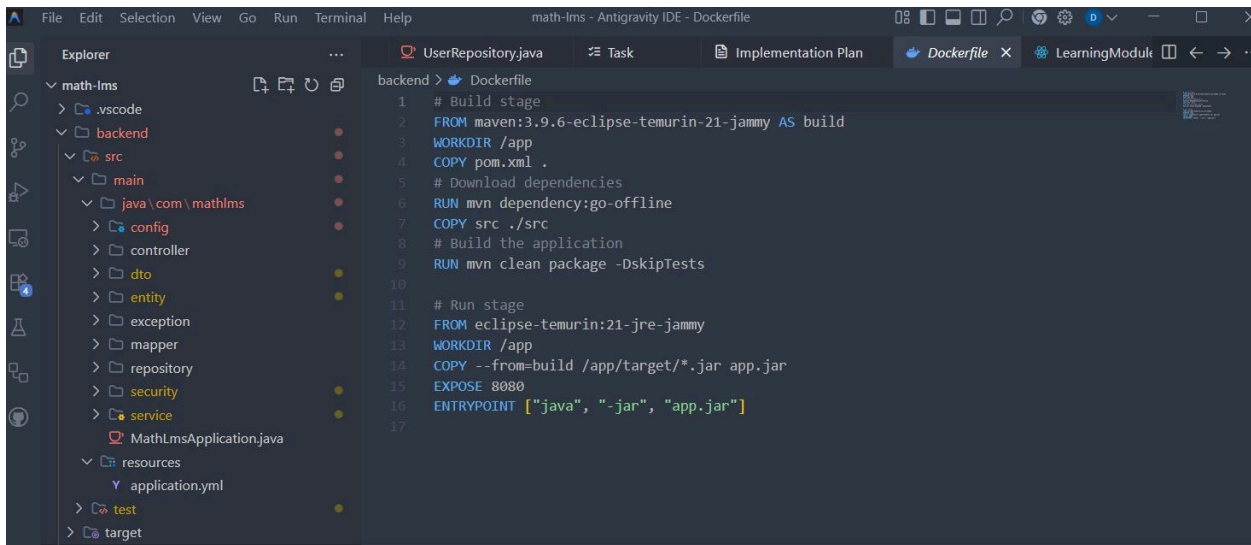
D dheepakadhithya@gmail.com  
Assigned recently

K kumar@gmail.com  
Assigned recently

J jagan@gmail.com  
Assigned recently

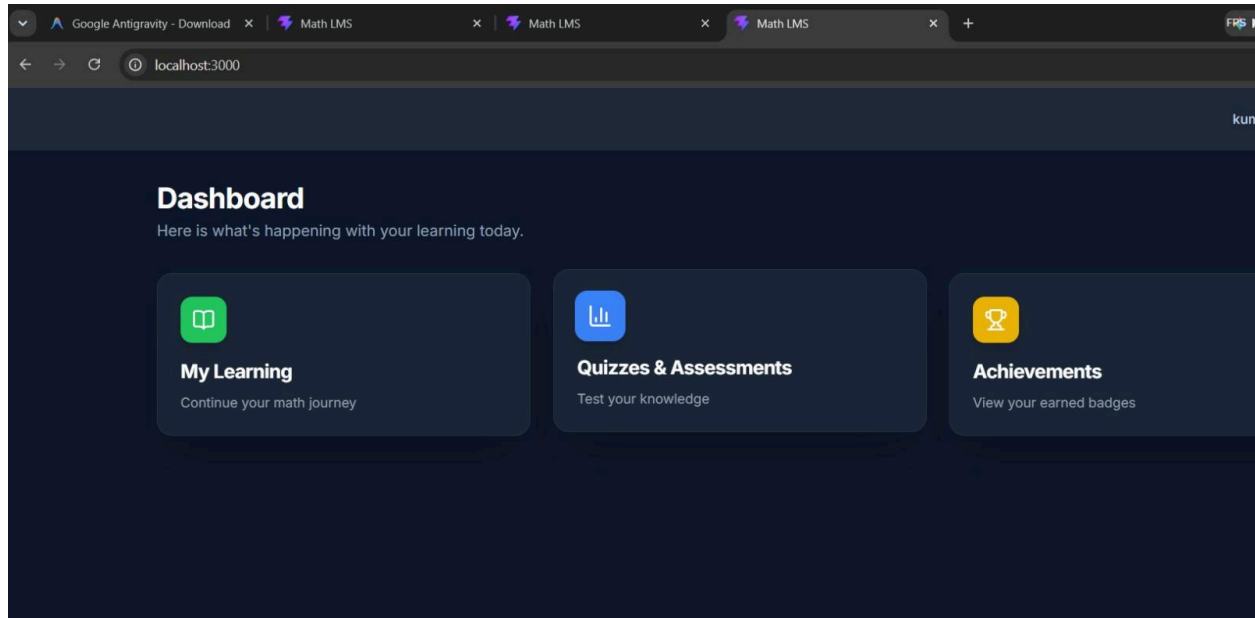
S sachin@gmail.com  
Assigned recently







# BANNARI AMMAN INSTITUTE OF TECHNOLOGY & INSTITUTION'S INNOVATION COUNCIL



### Customize Your Path

Select Grade

1

Select Topic

Counting

Difficulty Level

Easy

#### Video Tutorial



Numbers Song Let's Count 1-10 New Version  
Dream English Kids

Watch on YouTube  
Dream English Kids

#### Ready to Practice

Test your understanding level. AI will analyze your performance.

Questions: 10  
Estimated Time: 10 minutes

Start Practice

#### Theory & Documentation

Dive deep into the concepts and detailed documentation for Counting.

**Definition:**  
Determining the total number of objects in a set.

**Example:**  
1, 2, 3, 4...

#### Interactive Fun!







**BANNARI AMMAN INSTITUTE OF TECHNOLOGY  
&  
INSTITUTION'S INNOVATION COUNCIL**

